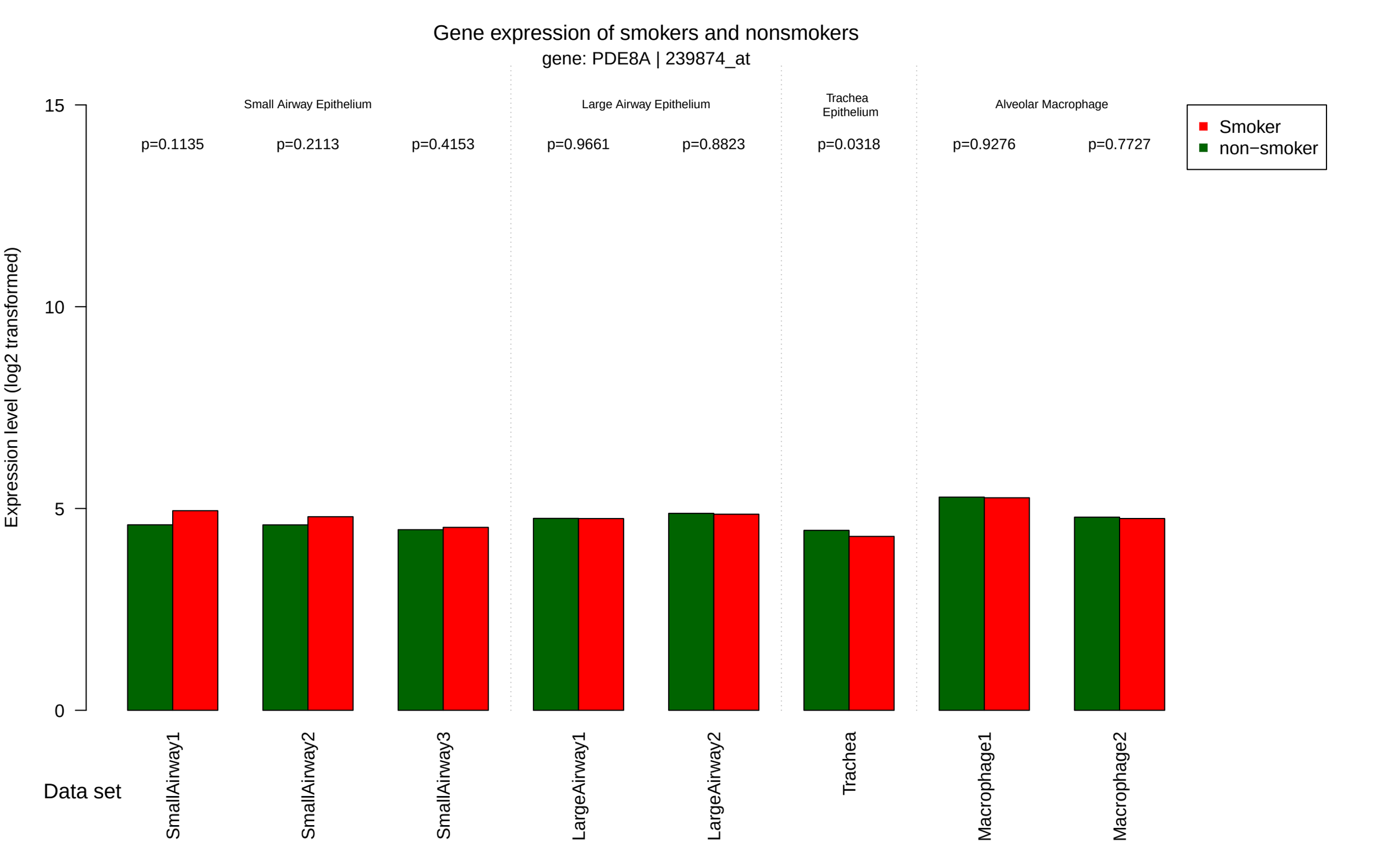
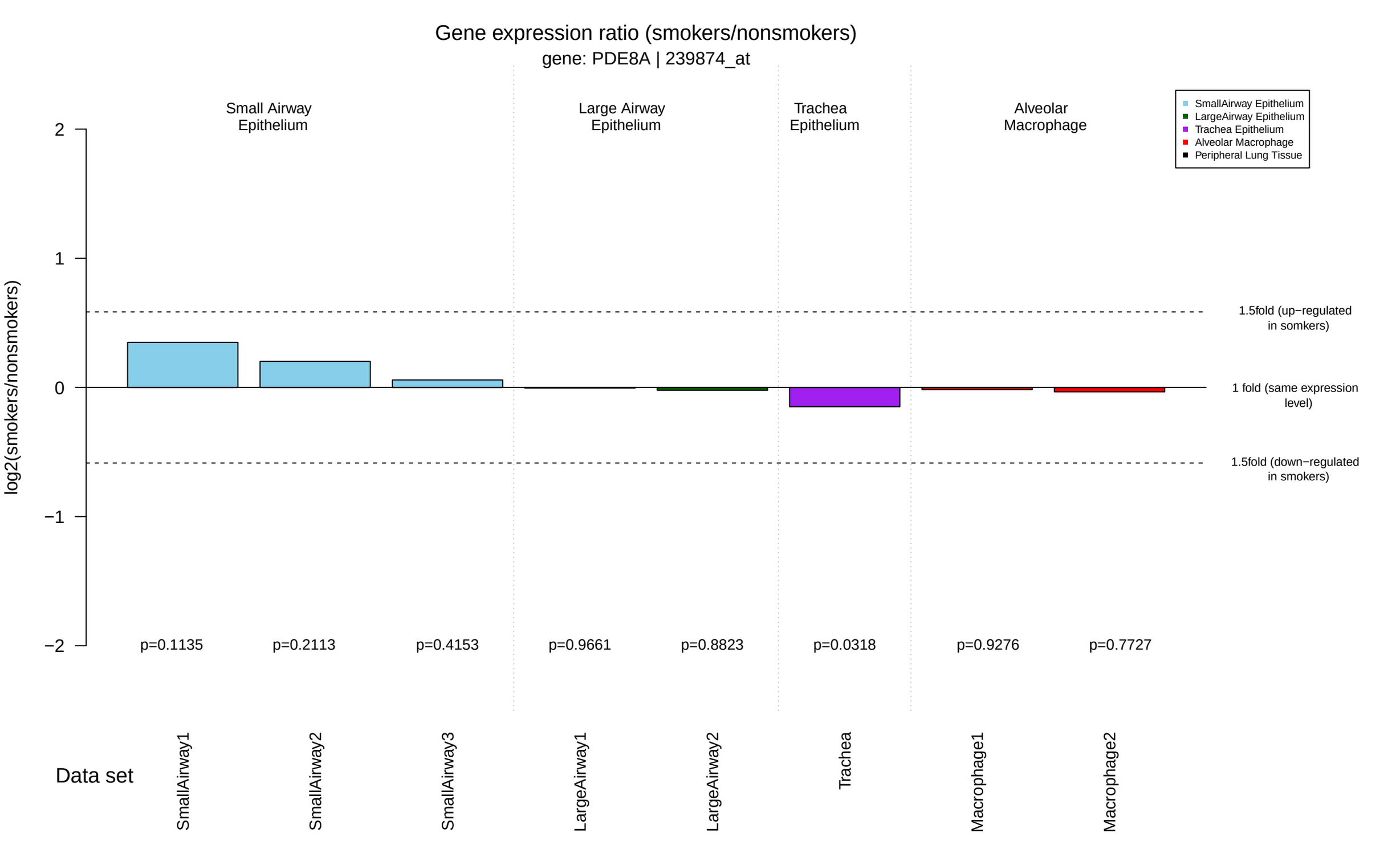
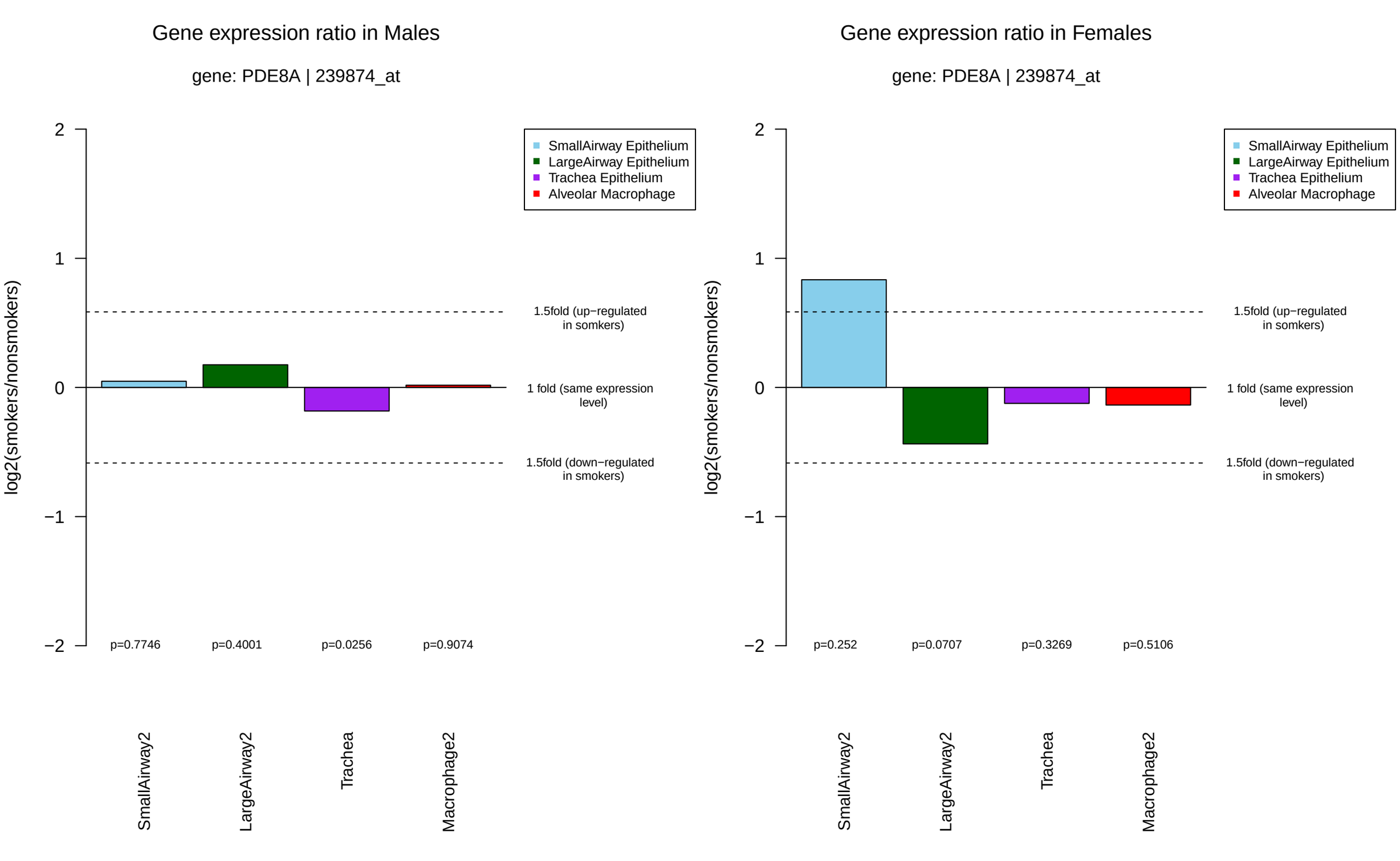
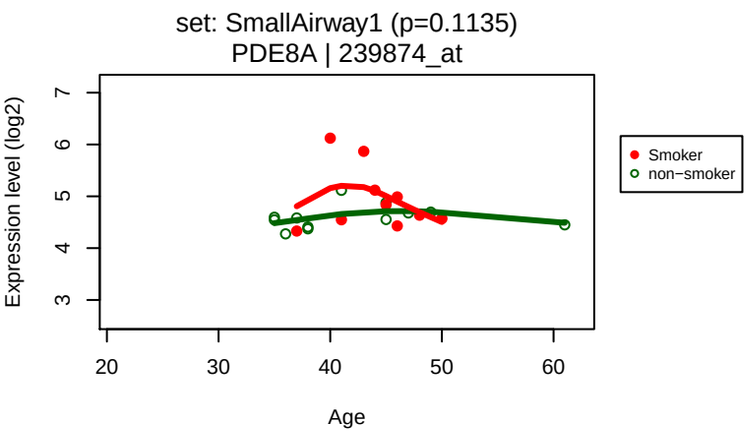
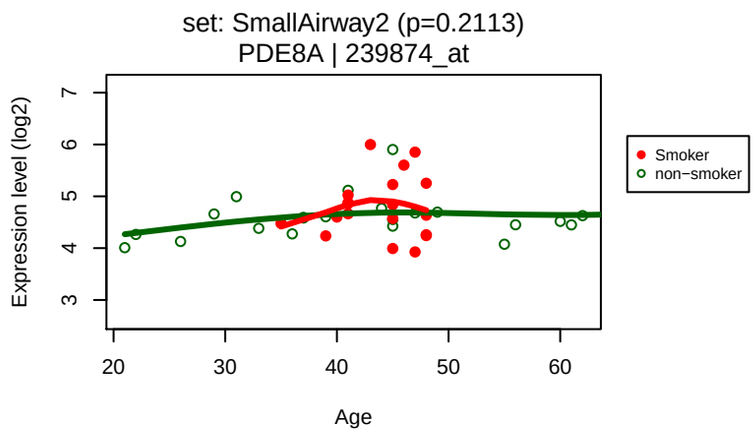
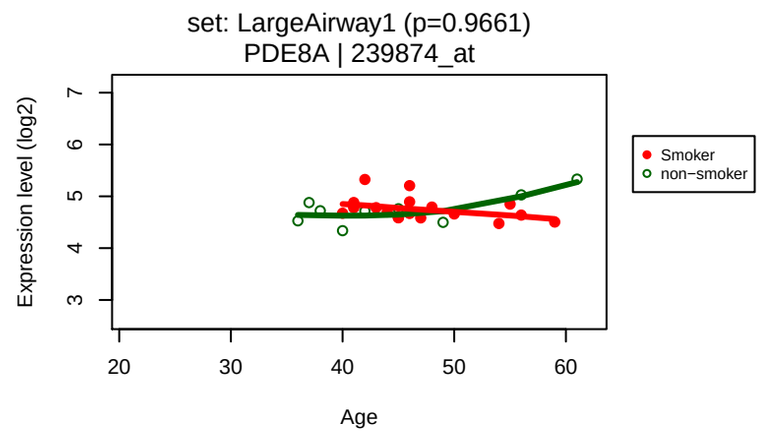

Gene expression of smokers and nonsmokers
gene: PDE8A | 239874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1135
p=0.2113
p=0.4153
p=0.9661
p=0.8823
p=0.0318
p=0.9276
p=0.7727
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

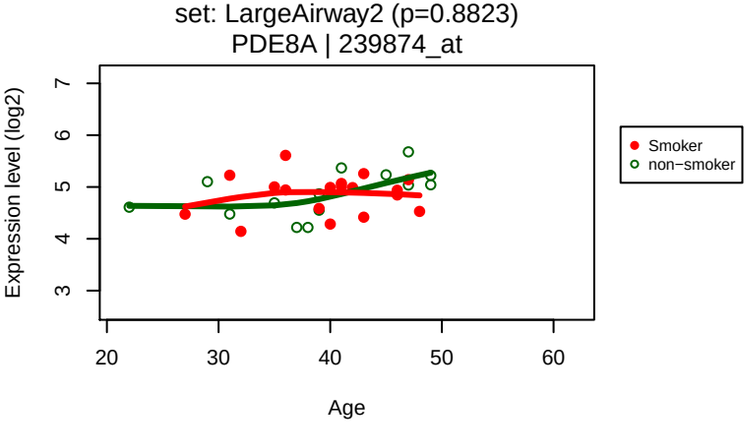

Gene expression ratio (smokers/nonsmokers)
gene: PDE8A | 239874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1135
p=0.2113
p=0.4153
p=0.9661
p=0.8823
p=0.0318
p=0.9276
p=0.7727
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

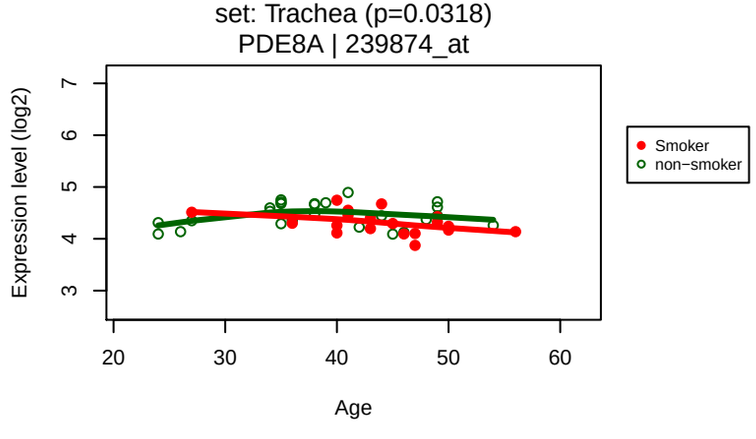

Gene expression ratio in Males
gene: PDE8A | 239874_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7746
p=0.4001
p=0.0256
p=0.9074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE8A | 239874_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.252
p=0.0707
p=0.3269
p=0.5106
SmallAirway2
LargeAirway2
Trachea
Macrophage2

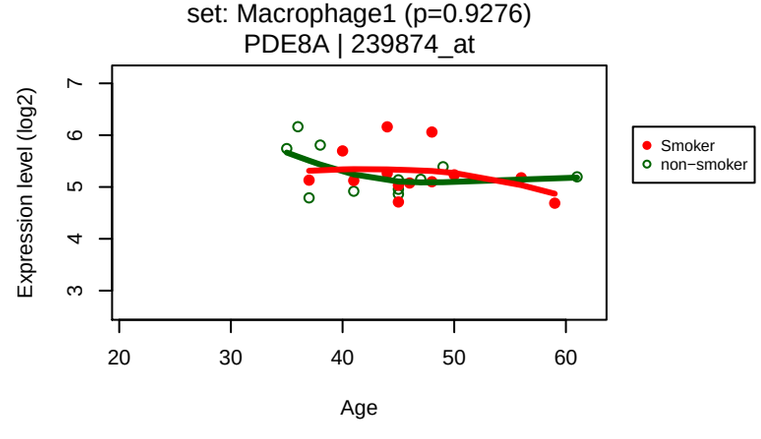

set: SmallAirway1 (p=0.1135)
PDE8A | 239874_at
Expression level (log2)
Age
Smoker
non−smoker

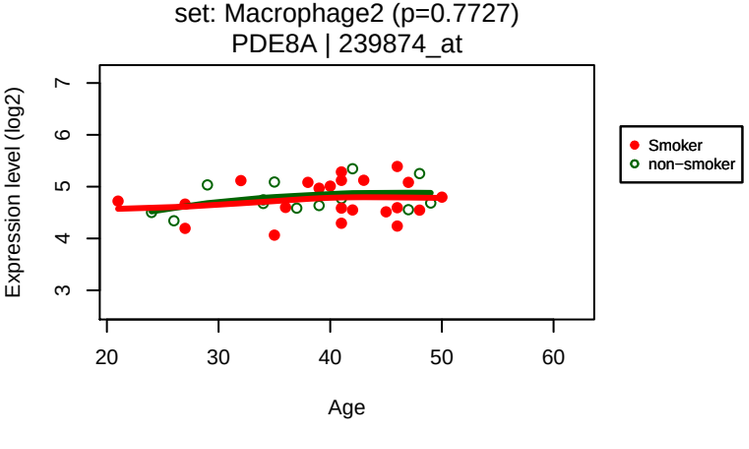

set: SmallAirway2 (p=0.2113)
PDE8A | 239874_at
Expression level (log2)
Age
Smoker
non−smoker

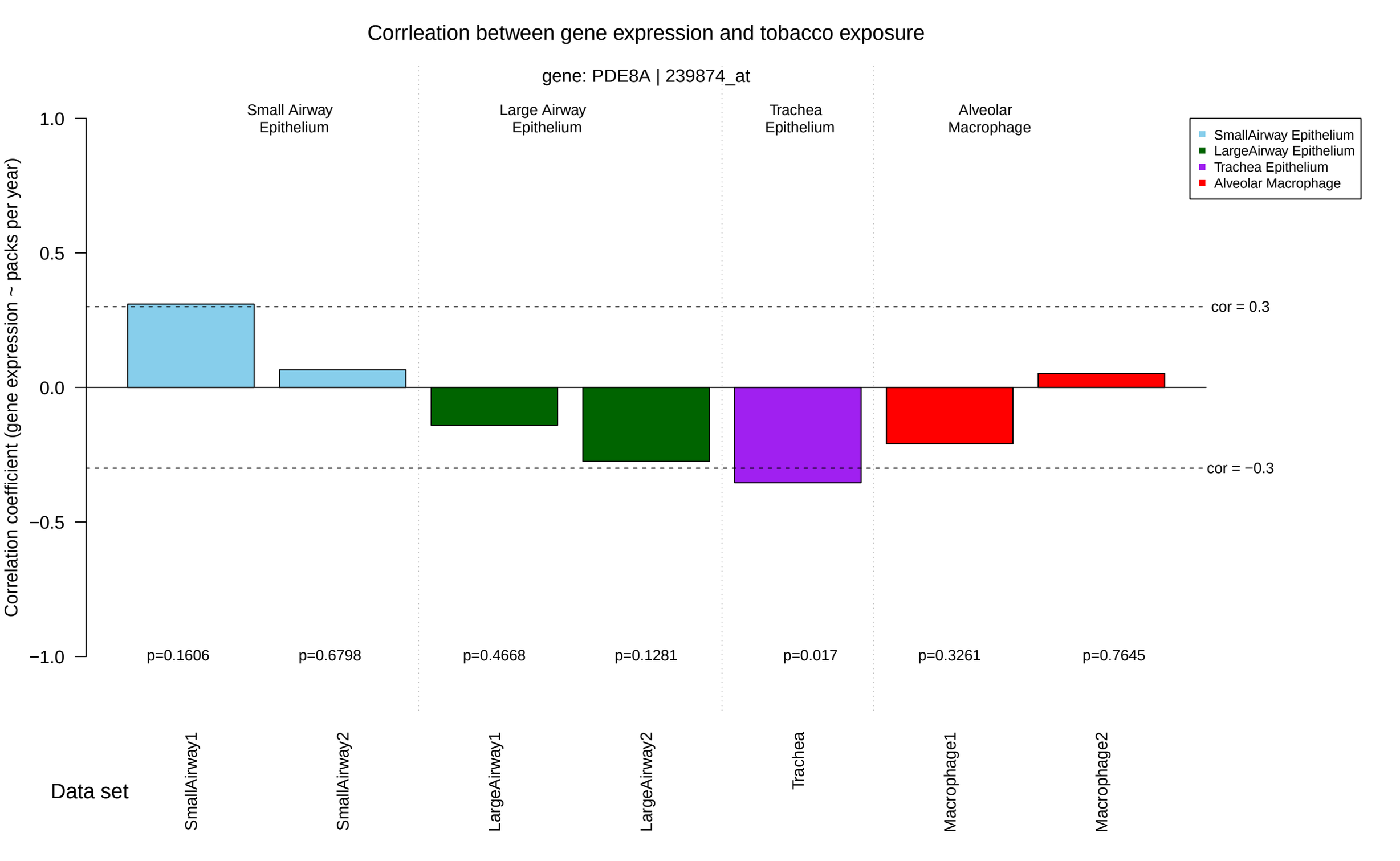

set: LargeAirway1 (p=0.9661)
PDE8A | 239874_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8823)
PDE8A | 239874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0318)
PDE8A | 239874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9276)
PDE8A | 239874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7727)
PDE8A | 239874_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE8A | 239874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1606
p=0.6798
p=0.4668
p=0.1281
p=0.017
p=0.3261
p=0.7645
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2